



AME 551| Designing Extended Reality Experiences

Monday | 1:30 PM - 4:15 PM

MIX 228

Instructors: Nicholas Pilarski & Tejaswi Gowda

<https://github.com/tejaswigowda/AME551-Spring25>

Office Location: MIX

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Office hours: By appointment.

Course Description

Augmented reality and virtual reality games, advertisements, installations, and other media integrate virtual objects with real-world environments in an emerging computing paradigm called *extended reality* (XR). Through this course, students will learn skills to creatively develop interactive mixed-reality objects and environments. Students will acquire skills to track and deploy *digital twins* of physical objects (both rigid bodies and rigged skeletons) in various 3D tools/environments. Integrating distributed networked computers and toolchains over the Internet including data interchange formats and protocols will be emphasized.

Enrollment Requirements

Not Applicable

Course Objectives

The goals of this course are to improve student familiarity with and proficiency in:

- (i) object creation and manipulation in virtual environments
- (ii) virtual physics and interactions,

- (iii) camera-projection coordinate systems,
- (iv) data interchange formats and network protocols,
- (v) Toolchaining for XR devops, and
- (vi) workflow processes for augmented/virtual-reality.

Although the mixed-reality, 3D modeling/visualization and networked programming principles will transcend programming language and environment, students will use C# (blueprint) programming in Unity and JavaScript for WebXR.

Student Learning Outcomes

At the end of this course, students will be able to:

- Demonstrate practical approaches to object-oriented programming
- Show evidence of a working knowledge of programming with a game engine, e.g., scripting interactions, physics, lighting, worldbuilding
- Demonstrate expertise in project conceptualization and development skills that lead to the construction of virtual environments and cinematics.
- Deploy, document, service and maintain such interdisciplinary projects.

Prerequisite: Students are expected to have a basic understanding of Unity or UE. Students can fulfill this prerequisite by either taking AME 230: "Programming for the Media Arts" or reviewing the Unity or Unreal coding materials.



Course Access

Your ASU courses can be accessed by both my.asu.edu and myasucourses.asu.edu; bookmark both if one site is down.

Additional Requirements

This course requires the following technologies.

- Access to Mesa Computer Lab or Laptop with a 2080 Graphics Card or Above
- USB C HD w/ 1 TB storage
- Smartphone
- Web browsers ([Chrome](#), [Mozilla Firefox](#), or [Safari](#))
- [Adobe Acrobat Reader](#) (free)
- [Adobe Flash Player](#) (free)
- Webcam, microphone, headset/earbuds, and speaker
- Microsoft Office ([Microsoft 365 is free](#) for all currently enrolled ASU students) • Reliable broadband internet connection (DSL or cable) to stream videos.

Classroom Technology (Your specific Technology Usage policy)

- It is encouraged that you bring technology (cell phones, tablets, and laptops) to class to help you take notes and do research, however, please turn off cell phone ringers and do not use your phone to make personal calls in class or use any technology to use social media in class. Do not answer your phone in class. If you believe you are receiving an emergency call, please step outside to take it.

Student Success and the Classroom Learning Community

To be successful:

You are part of the learning community in this course and expected to engage with the material, discussion and activities with care and integrity, and to respect the rights of others in carrying out all academic assignments and classroom discussions. We expect active and informed participation in our class discussions, activities and creative practices. Part of the responsibility for that community learning is the health and wellbeing of others. We will follow all ASU policies and recommendations.

Additional Success Strategies

- check the course daily
- read announcements
- read and respond to course email messages as needed
- complete assignments by the due dates specified
- communicate regularly with your instructor and peers
- create a study and/or assignment schedule to stay on track
- access [ASU Student Resources](#)



Required Primary and Secondary Materials (e.g., readings, videos, podcasts, films, and studio supplies)

Texts:

Brown, Blain. Cinematography: Theory & Practice. 3rd ed., Routledge,

2016. ABOVE TEXT IS OPTIONAL

*All additional texts will be made available for download.

Software:

UE5 (Free)

Please budget 50 to 100 USD for potential UE5 Assets and Toolkits

Supplies:

Computer with proper minimum specs for Unreal and Blender (computer lab access provided)

External SD 1tb USB-C.



Anti-Discrimination Statement

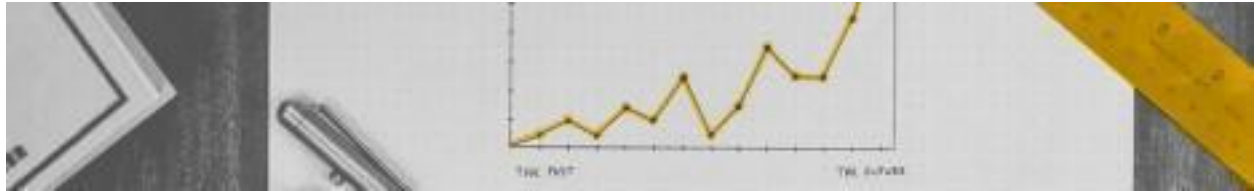
The Herberger Institute of Design and the Arts at Arizona State University upholds, values, and cherishes student and faculty diversity, no matter the circumstance. As members of the ASU community, we are charged with challenging injustices and social inequities of any kind through education. These values are an integral part of our standing as an institution and must be upheld by all members of the ASU community, including but not limited to all Herberger Institute of Design and the Arts staff, faculty and students. The call is clear and present at ASU for every member of our community to do their part in fostering a culture of Inclusive Excellence that contributes meaningfully to lasting equity for all. For students and faculty alike, this culture of Inclusive Excellence creates role models, broadens perspectives, combats negative stereotyping and enables artists, designers and makers of the 21st century to think creatively, critically and, above all, compassionately about our impact on the world at large.

This course and Arizona State University welcomes all students regardless of race/ethnicity, gender identities, gender expressions, sexual orientation, socio-economic status, age, disabilities, religion, regional background, Veteran status, citizenship status, nationality and other diverse identities that we each bring to class. Each of us bear intersectional perspectives that are born out of our backgrounds and identities, and each of us has a contribution to make towards our culture of Inclusive Excellence. As your instructor, I expect that all of my students abide by the following community agreements:

- to bring a willingness to deeply inspect your own assumptions about the world, identifying areas in which you may need to unlearn implicit biases and behaviors
- to help others learn by respectfully voicing your thoughts and reactions, acknowledging that they are partial to and shaped by the way you make sense of the world
- to demonstrate a curious and eager inquiry into how others make sense of the world

Diversity of experiences, backgrounds and opinions are essential to cultivating a rich academic environment which in turn strengthens our capacity to be ethical and empathetic creative-thinkers. I aim for students of all backgrounds to be well served by this course and that the diversity students bring to this class be viewed as a resource, strength and benefit. The materials of this course, including readings and assignments, will reflect this commitment to diverse and inclusive knowledge. If a student feels that there has been a discrepancy between my teaching practices and the university's commitment to Inclusive Excellence, you are encouraged to discuss your concerns directly with me. I value your learning experience and welcome all opportunities to enrich the efficacy of this course for all student groups. Here are resources available to all students to report incidents of bias, harassment, and other forms of discrimination inside and outside the classroom:

- Unsure of whether the concern you experience or witness falls under the criteria of bias, harassment, and other forms of discrimination? You can fill out the Herberger Institute Community of Care form at herbergerinstitute.asu.edu/caring describing the situation. One of the members of the Herberger Institute Behavioral Response Team will connect with you.
- Anyone who believes that they have been subjected to discrimination, harassment, or retaliation in violation of this policy, or who believes that this policy has been violated, should report the matter immediately to the Office of University Rights and Responsibilities or the Dean of Students office or directly fill out an incident report.
- Unless a person is restricted by law from doing so, any employee who is informed of or has a reasonable basis to believe that sexual harassment has occurred, shall immediately report all information regarding the occurrence(s) to the Office of University Rights and Responsibilities or the Title IX Coordinator or the Dean of Students office.
- More reporting pathways are available to students on the University's Prohibition Against Discrimination, Harassment, and Retaliation policy page.



Assignments

Attendance, Participation, and In-class Exercises. (30 points total)

You are required to be active and participate in all coursework and discussion.

Absences and participation reflect your overall commitment to advancing your own understanding of concepts and skills ascertained in the course. In addition your efforts add to the culture of the classroom. Active participation and informed understanding of the assigned readings/modules comprise a large percentage of this grade. Please come with talking points, completed modules and be prepared to be called on directly and/or show work.

Due to the potential flux of “active participation and informed understanding,” if you are ever curious of where you stand, please do not hesitate to reach out to all professors in a single email to request where you currently stand in this regard.

Micro-projects (4 projects, 0-10 points awarded for each project, 40 points total)

You will focus your efforts on four small projects which are designed to help you advance XR design and interaction.

Projects will be graded as follows:

10 points: Demonstrates originality in production, critical thinking, and deep engagement with the formal qualities of interactive XR informed by tutorials/lectures/readings. In addition to this, all programming in the virtual environment simply works!

5 points: Demonstrates effort to engage with the films/lectures/readings but does not reflect advanced critical thought and/or craft. Some object oriented problems which require attention.

0 points: Shows clear lack of effort or failure to complete assignment.

Late Assignments: Students will be given five late days for use on the assignments; at most two can be used for any one assignment. The purpose of late days is to allow students the flexibility to manage unexpected obstacles to coursework that arise during the course of the term, such as travel, moderate illness, conflicts with other courses, extracurricular obligations, job interviews, etc. Thus, additional late days will NOT be granted except under truly exceptional circumstances.

- Handing in an assignment at 8:15am the morning after it is due consumes one late day.
- Handing in an assignment at 4:30pm the day after it is due consumes one late day.
- Handing in an assignment at 3:30am two days after it is due consumes two late days.

Projects must be exported/uploaded/available by start-of-class on the due date/critique. Please test your project before presenting. Three points will be deducted from projects received later than start-of-class on due-date or those that are “broken” or fail to run. Two point will be deducted for each additional day the work is late.

Make-Up Assignments (Max 2)

If you do not like the grade you receive on the assignment you may return your assignment for a new grade within 5 days. You can do this 2 times, however do not have this option for the final. Along with the resubmission you must provide a memo outlining what you changed and what you learned from the revision process.

Final Project (30 points)

All students will be required to produce a “final work” with accompanying artist statement. The work of art must be informed by the tutorials, readings, and/or concepts covered in class. Works can be original or born through iterative design that is built into the curriculum.

Detailed information and requirements will be provided later in the semester and expectations should become self-evident through coursework and lectures.

No late assignments will be accepted for final!

Grading, including grade scale

Attendance/Participation/Modules 40 points

Micro Project 1 – Pitch Final 10 points

Micro Project 2 - Environment 10 points

Micro Project 3 – Blueprints 01 10 points

Micro Project 4 – Blueprints 02 10 points

Final Project, Analysis, and Design 30 points

Total 100 points

Grade Scale

98+	A+
93-97	A
90-92	A-
88-89	B+
83-87	B
80-82	B-
78-79	C+
70-77	C
60-69	D
0-59	E

Rounding Policy

Under no circumstances do we round grades.

Y GRADE

The Y (Satisfactory) grade is an option for all classes and does not impact DARS requirements, although it is NOT included in GPA. If students will have the option for a Y grade in this course, then please include the conditions and processes they should follow to request a Y grade.

Incomplete Policy

incomplete is not a routine process for successful completion of coursework. Rather, it is a limited academic exception intended to address situations where a student who has been doing acceptable work experiences exceptional extenuating circumstances beyond the student's control preventing their timely completion of the course. In evaluating requests for incompletes, I not only assess the nature of the extenuating circumstances and whether such circumstances were beyond the student's control, but also whether the student's record of performance in the class demonstrates the likelihood for successful completion of the remaining coursework. All incomplete contracts must also be approved from the academic unit leads. Finally, while policy dictates the maximum time for contract completion, I take into account what materials are outstanding and consider the appropriate extension. Statistically at ASU, the shorter the contract length the more likely students will be able to successfully complete their classroom obligations. You can read more about the incomplete policy at <https://www.asu.edu/aad/manuals/ssm/ssm203-09.html>

Subject to change

The Instructor reserves the right to change portions of this syllabus (assignments, lectures, deadlines etc.) by verbal instructions during scheduled class time. The student is responsible for noting changes and acting accordingly. If the student is absent from class, they are responsible to ask fellow classmates. Grading and absence policies are not subject to change.



Course Itinerary (Schedule)

*Please note that the following schedule is subject to change.

Week	Date	XR Topic	Theory Topic	Due
1	01/13	Class Intro		
2	01/20	No Class MLK Day		
3	01/27	XR and Unreal Intro - Design and XR as a Form	What is New Media	
4	02/03	Project Pitches	Ethics of simulated architecture vs embodied reality.	Micro Project 1: Pitch
5	02/10	WorldBuilding - Ingestion and Assembly - Static Meshes and Lighting 1	AI in Game Engines	
6	02/17	World Building - Foliage and Materials		
7	02/24	Blueprints 01 (Triggers - Sonic)	Cultural responsibility of the programmer in simulated worlds	Micro Project 2: World
8	03/03	Digital Twins: mapping orientation and position		
9	03/10	SPRING BREAK		
10	03/17	Blueprints (Custom Project Workshop)		
11	03/24	Intro to Animation on the Timeline + Cinematics, scenes, etc (Blueprints Animations)	Economics of XR - An economy of access.	
12	03/31	Intro to the Oculus and Setting up Environment	Ethics of simulated humans.	Micro Project 3 Prefab Project
13	04/07	Blueprint Buttons, and UI, realtime mirroring of rigid bodies		
14	04/14	Real time networked reality capture	Cinema of attraction - false realities and counter-cultural uses of XR.	Micro Project 4: Individual Project
15	04/21	Workshop - Usability	Critique	

		UIUX 02		
16	04/28	Final	Critique	Final: XR Experience



Policies and Procedures

Attendance Policy (see:

<https://provost.asu.edu/fall-2022-learning-expectations> for additional language)

Attendance and participation for the duration of the class period is mandatory. If you have more than 3 absences (unexcused), your final grade will be lowered 1 grade for each subsequent absence (i.e. B to C). You should notify me by email prior to absence if possible and provide doctor's note where applicable. Repeated tardiness and leaving class early will be recorded, and as a result, your final grade will be lowered. It is the student's responsibility to keep track of his/her absences.

Attendance and participation in class activities is an essential part of the learning process, and students are expected to attend class regularly. Some absences are, however, unavoidable. Excused absences for classes will be given without penalty to the grade in the case of (1) a university sanctioned event [ACD 304-02]; (2) religious holidays [ACD 304-04; a list can be found [here https://eooss.asu.edu/cora/holidays](https://eooss.asu.edu/cora/holidays)]; (3) work performed in the line-of-duty according [SSM 20118]; and (4) illness, quarantine or self-isolation related to illness as documented by a health professional.

Anticipated absences for university-sanctioned events, religious holidays, or line-of-duty activity should be communicated to the instructor by **email** at least **48 hours** before the expected absence.

Absences for illness, quarantine or self-isolation related to illness should be documented by a health professional and communicated to the instructor as soon as possible via email with appropriate attachments.

Excused absences do not relieve students from responsibility for any part of the course work required during the period of absence. Faculty will provide accommodations that may include participation in classes remotely, access to recordings of class activities, and make-up work.

Excused absences related to religious observances/practices in accord with [ACD 304-04](#), "Accommodation for Religious Practices." Students may be excused for the observance of religious holidays. Students should notify the instructor at the beginning of the semester about the need to be absent from class due to religious observances. Students will be responsible for materials covered during their absence and should consult with the instructor to arrange reasonable accommodation for missed exams or other required assignments.

Excused absences related to university sanctioned activities in accord with [ACD 304-02](#), "Missed Classes Due to University-Sanctioned Activities." Students required to miss classes due to university sanctioned activities will not be counted absent. However, absence from class or examinations due to university-sanctioned activities does not relieve students from responsibility for any part of the course work required during the period of the absence. Students should inform the instructor early in the semester of upcoming scheduled absences and immediately upon learning of unscheduled required class absences. Reasonable accommodation to make up missed exams or other required assignments will be made. Consult the instructor BEFORE the absence to arrange for this accommodation.

Line-of-duty absence and missed assignment policy

A student who is a member of the National Guard, Reserve, or other U.S. Armed Forces branch who misses classes, assignments or examinations due to line-of-duty responsibilities shall have the opportunity to make up the coursework in accordance with [SSM 201-18 Accommodating Active Duty Military Personnel](#). This accommodation also applies to spouses who are the guardian of minor children during line-of-duty activities. This policy does not excuse students from course responsibilities during their absence. Students should first notify the Pat Tillman Veterans Center of their activation and then the instructor to discuss options.

Instructor Absence Policy

Students should wait for an absent instructor 15 minutes in class sessions of 90 minutes or less, and 30 minutes for those lasting more than 90 minutes, unless directed otherwise by someone from the academic unit.

Academic Integrity and Student Honor Code

The ASU student honor code affirms the commitment of ASU to uphold the values, principles, and ethics of academic integrity. All students are expected follow the code which states,

"We, the students of Arizona State University, have adopted this code as an affirmation of our commitment to academic integrity and our participation in ethical education. We embrace the duty to uphold ASU's Honor Code, and in light of that duty, We promise to refrain from academic dishonesty. We pledge to act with integrity and honesty to promote these values among our peers. We agree to always abide by the Sun Devil Way and uphold the values of the New American University."

Every student is expected to produce his/her original, independent work. Any student whose work indicates a violation of the ASU Academic Integrity Policy including cheating, plagiarism, and dishonesty will be subject to disciplinary action. Plagiarism is defined as deliberately passing off someone else's words or ideas as your own. All necessary and appropriate sanctions will be issued to all parties involved with plagiarizing any and all course work. Plagiarism and any other form of academic dishonesty that is in violation with the Student Code of Conduct will not be tolerated. Arizona State University and the Herberger Institute for Design and the Arts expect the highest standards of academic integrity from all students. Failure to meet these standards may result in suspension or expulsion from the university or other sanctions as specified in the ASU Student Academic Integrity Policy (<http://provost.asu.edu/academicintegrity>), "[e]ach student must act with honesty and integrity, and must respect the rights of others in carrying out all academic assignments." This policy also defines academic dishonesty and sets a process for faculty members and colleges to sanction dishonesty. Violations of this policy fall into five broad areas that include but are not limited to:

- Cheating on an academic evaluation or assignments
- Plagiarizing
- Academic deceit, such as fabricating data or information
- Aiding Academic Integrity Policy violations and inappropriately collaborating
- Falsifying academic records

I sanction any incidents of academic dishonesty in my courses using University and HIDA guidelines. Should you have any question about whether or not something falls subject to this clause, feel free to contact me or review the university policy on academic integrity at the above link. Per ASU policy, a student may not avoid the consequences of academic dishonesty by withdrawing from a course, and may be placed back in the course in order to face sanctions resulting from academic integrity violations. You are responsible for abiding by this policy.

Student Learning Community Conduct

ASU adheres to a university-wide Student Code of Conduct. The philosophy behind this policy states, The aim of education is the intellectual, personal, social, and ethical development of the individual. The educational process is ideally conducted in an environment that encourages reasoned discourse, intellectual honesty, openness to constructive change and respect for the rights of all individuals. Self discipline and a respect for the rights of others in the university community are necessary for the fulfillment of such goals. The Student Code of Conduct is designed to promote this environment at each of the state universities. You are expected to treat your instructor and your fellow classmates with respect and kindness. In all correspondence and in Discussion Board postings, you should show respect for the viewpoints of others who may disagree with you or see things from a different perspective. Criticizing, ridiculing, insulting, or belittling others will not be accepted. Keep in mind that electronic communications do not have the advantage of nonverbal cues that are so much a part of interpersonal communication. Humor or satire can sometimes be misinterpreted in strictly electronic communication forums. In addition, all students should be aware of their [Rights and Responsibilities](#) at Arizona State University.

Copyright

Students must refrain from uploading to any course shell, discussion board, or website used by the course instructor or other course forum, material that is not the student's original work, unless the students first comply with all applicable copyright laws; faculty members reserve the right to delete materials on the grounds of suspected copyright infringement. The course content, including lectures and other handouts, is copyrighted material. Students may not share outside the class, upload, sell, or distribute course content or notes taken during the conduct of the course (see [ACD 304-06](#), "Commercial Note Taking Services" for more information). THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, SOLD, OR DISTRIBUTED.

Statement on ASU's Community of Care standards

The Herberger Institute for Design and the Arts complies with the spirit and the letter of ASU's community of care standards with regard to social distancing, masking, and student, faculty, and staff safety and well being. <https://eooss.asu.edu/communityofcare>

Threatening or disruptive behavior

Self-discipline and a respect for the rights of others in the classroom or studio and university community are necessary for a conducive learning and teaching environment. Threatening or violent behavior will result in the administrative withdrawal of the student from the class. Disruptive behavior may result in the removal of the student from the class. Threatening, violent, or disruptive behavior will not be tolerated in this class, and will be handled in accordance with ASU policy (SSM 104-02). For more information please visit:

<https://eooss.asu.edu/dos/srr/PoliciesAndProcedures> and
<https://eooss.asu.edu/dos/safety/ThreateningBehavior>.

Withdrawal

If you are unable to complete the course, it is your responsibility to arrange for withdrawal from the class. You will not be automatically withdrawn and unless you are officially withdrawn from the course you will receive a final grade based upon the total points you have earned for the semester. Students are required to pay all tuition and fees for any registered course unless enrollment is officially cancelled during the 100% refund period. Please visit the Academic Calendar to review the withdrawal deadlines for this semester. For more information on Drop/Add and Withdrawal visit <https://students.asu.edu/drop-add>

Special Accommodations

Your instructor will make any reasonable adaptations for limitations due to any disability documented with the Student Accessibility and Inclusive Learning Services (SAILS), including learning disabilities.

Please contact the instructor during office hours or by appointment to discuss any special needs you may have. You must contact the SAILS to process the paperwork for special course accommodations.

To request academic accommodations due to a disability, please contact the SAILS (<https://eooss.asu.edu/drc>); Phone (480) 965-1234; TDD (480) 965-9000). Students who feel they will need disability accommodations in this class but have not registered SAILS should contact SAILS immediately. Students should contact the Center on the campus that your class is being held. Campus-specific [location and contact information](#) can be found on the SAILS website. SAILS offices are open 8 a.m. to 5 p.m. Monday – Friday. Check the [above](#) website for eligibility and documentation policies (<https://eooss.asu.edu/drc>). This is a very important step as accommodations may be difficult to make retroactively. If you have a letter from their office indicating that you have a disability which requires academic accommodations, in order to assure that you receive your accommodations in a timely manner, please present this documentation to me as soon as possible so that your needs can be addressed effectively.

Title IX and Mandated Reporter Policy

Title IX is a federal law that provides that no person be excluded on the basis of sex from participation in, be denied benefits of, or be subjected to discrimination under any education program or activity. Both Title IX and university policy make clear that sexual violence and harassment based on sex is prohibited. An individual who believes they have been subjected to sexual violence or harassed on the basis of sex can seek support, including counseling and academic support, from the university. If you or someone you know has been harassed on the basis of sex or sexually assaulted, you can find information and resources at <https://sexualviolenceprevention.asu.edu/faqs>.

As a mandated reporter, I am obligated to report any information I become aware of regarding alleged acts of sexual discrimination, including sexual violence and dating violence. ASU Counseling Services, <https://eooss.asu.edu/counseling>, is available if you wish to discuss any concerns confidentially and privately.

Policy on Sexual Discrimination

Policy on sexual discrimination as described in [ACD 401](#), "Prohibition Against Discrimination, Harassment, and Retaliation", including the fact that the instructor is a mandated reporter and therefore obligated to report any information regarding alleged acts of sexual discrimination. Arizona State University is committed to providing an environment free of discrimination, harassment, or retaliation for the entire university community, including all students, faculty members, staff employees, and guests. ASU expressly prohibits [discrimination](#), [harassment](#), and [retaliation](#) by employees, students, contractors, or agents of the university based on any protected status: race, color, religion, sex, national origin, age, disability, veteran status, sexual orientation, gender identity, and genetic information.

Student Services & Resources

You will find a list of student resources at <https://eooss.asu.edu/resources>

Resources included are advisement, registration, financial aid, disability services, counseling, tutoring, library, and more.

Novel Coronavirus Information and Updates

You will find information and Frequently Asked Questions here <https://eooss.asu.edu/communityofcare>

You will find Novel Coronavirus updates and announcements here <https://eooss.asu.edu/health/announcements/coronavirus>

Non-emergency Student Care process

If you are concerned for a your own or a fellow student's well-being, please review the information and complete the form at herbergerinstitute.asu.edu/caring and the HIDA Care Team will reach out. FOR EMERGENCIES CALL 911. (Be prepared with the physical address of the location.)

Academic Calendar and Important Dates

The academic calendar can be found here <https://students.asu.edu/academic-calendar>